

Smart contract audit

rain.math.binary

Content

Project description	3
Executive summary	3
Reviews	3
Technical analysis and findings	4
Security findings	5
**** Critical	5
*** High	5
** Medium	5
* Low	5
Informational	5
General Risks	6
Approach and methodology	7



Project description

This repository contains LibCtPop, a specialized Solidity library designed for high-performance bitwise operations. Its primary function, `ctpop` (count population), calculates the Hamming weight—the number of one-bits—in a `uint256` value.

The library provides two implementations: an optimized version utilizing parallel bit counting strategies and magic constants for maximum gas efficiency, and a reference implementation (`ctpopSlow`) for verification purposes. These tools are intended for use cases requiring efficient low-level bit manipulation and mathematical computations on the Ethereum Virtual Machine (EVM).

Executive summary

Type	Library
Languages	Solidity
Methods	Architecture Review, Manual Review, Unit Testing, Functional Testing, Automated Review
Documentation	README.md
Repositories	https://github.com/rainlanguage/rain.math.binary

Reviews

Date	Repository	Commit
26/01/26	rain.math.binary	c7ebb6cb99e1648d18594f888331bd584c6a911d

Scope

Contracts
src/lib/LibCtPop.sol



Technical analysis and findings

Critical	0
High	0
Medium	0
Low	0
Informational	0



Security findings

**** Critical

No critical severity issue found.

*** High

No high severity issue found.

** Medium

No medium severity issue found.

* Low

No low severity issue found.

Informational

No informational severity issue found.



General Risks

- Edge Case Reliance: The optimization logic requires explicit handling of specific edge cases (e.g., `type(uint256).max`) that fall outside the general algorithm's coverage. Ensuring all such edge cases are identified and handled is critical for maintaining correctness across the full input range.



Approach and methodology

To establish a uniform evaluation, we define the following terminology in accordance with the OWASP Risk Rating

Methodology:

	Likelihood Indicates the probability of a specific vulnerability being discovered and exploited in real-world scenarios
	Impact Measures the technical loss and business repercussions resulting from a successful attack
	Severity Reflects the comprehensive magnitude of the risk, combining both the probability of occurrence (likelihood) and the extent of potential consequences (impact)

Likelihood and impact are divided into three levels: High H, Medium M, and Low L. The severity of a risk is a blend of these two factors, leading to its classification into one of four tiers: Critical, High, Medium, or Low.

When we identify an issue, our approach may include deploying contracts on our private testnet for validation through testing. Where necessary, we might also create a Proof of Concept PoC to demonstrate potential exploitability. In particular, we perform the audit according to the following procedure:

	Advanced DeFi Scrutiny We further review business logics, examine system operations, and place DeFi-related aspects under scrutiny to uncover possible pitfalls and/or bugs
	Semantic Consistency Checks We then manually check the logic of implemented smart contracts and compare with the description in the white paper.
	Security Analysis The process begins with a comprehensive examination of the system to gain a deep understanding of its internal mechanisms, identifying any irregularities and potential weak spots.